

TANVIR AHMED

AI Engineer | Machine Learning Engineer | Generative AI

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PROFESSIONAL SUMMARY

AI Engineer specializing in designing and deploying end-to-end AI systems across Machine Learning, Computer Vision, and Generative AI. Experienced in building scalable ML and AI pipelines using Python, PyTorch, TensorFlow, FastAPI, Docker, and AWS. Hands-on experience developing medical AI applications, predictive models, RAG architectures, and multi-agent systems, with knowledge of MLOps practices including DVC, MLflow, CI/CD, and cloud deployment.

Technical Projects & Experience

- Machine Learning & Deep Learning: Supervised Learning, Unsupervised Learning, PyTorch, TensorFlow, Transfer Learning, YOLO, OpenCV, Scikit-learn, XGBoost, NumPy, Pandas, Seaborn
- Generative AI & Agentic Frameworks: LangChain, LangGraph, OpenAI API, Gemini, CrewAI, Retrieval-Augmented Generation (RAG), Multi-Agent Systems
- MLOps & Cloud Deployment: MLflow, Dagshub, DVC, Docker, FastAPI, GitHub Actions, CI/CD, AWS (EC2, S3), Google Cloud Platform (GCP), Git, Linux, Bash, n8n
- Databases & Vector Search: SQL, MongoDB, Pinecone, ChromaDB

TECHNICAL PROJECTS & EXPERIENCE

Multi-Modal Retrieval-Augmented Generation (RAG) System - 🔗 [GitHub](#)

Role: AI Engineer | Tech: OpenAI Vision, Pinecone, Python, LangChain, AWS

- Architected a Multimodal RAG system capable of extracting and reasoning over text, complex tables, and embedded images inside unstructured PDF documents.
- Built a vision-aware semantic search pipeline using Pinecone and vision-LLMs to preserve document layout context.
- Reduced retrieval hallucinations by 25% using a hybrid retrieval strategy combining vector embeddings with cross-modal image-to-text descriptions.

AI Medical Assistant (LLM & RAG Healthcare System) - 🔗 [GitHub](#)

Role: AI Engineer | Tech: LangChain, OpenAI API / Gemini, Pinecone, FastAPI, Python

- Architected an end-to-end medical query assistant leveraging Retrieval-Augmented Generation (RAG) to provide contextual, evidence-backed healthcare insights.
- Implemented semantic search with vector embeddings in Pinecone to accurately match patient queries against medical datasets while mitigating hallucinations.
- Built robust safety guardrails and prompt templates to format responses for patient accessibility and ensure safe AI-driven medical communication.
- Developed a fast REST API endpoint using FastAPI to serve low-latency inference for real-time user interactions.

NeuroVision AI: Brain Tumor Detection, Segmentation & RAG Assistant - 🔗 [GitHub](#)

Role: Lead Developer | Tech: TensorFlow, Transfer Learning, YOLO, SAM, FastAPI, Pinecone, AWS, OpenAI

- Built a multi-stage clinical AI pipeline for MRI analysis, combining VGG16 for classification, YOLO for localization, and Segment Anything Model (SAM) for precise tumor boundary extraction.
- Developed an AI-powered medical chatbot using RAG architecture, Pinecone vector database, and Large Language Models (LLMs) deployed on AWS.
- Deployed a full-stack application using FastAPI to enable real-time inference and visualization of segmented tumor masks for clinical review.

PneumoScan-AI Automated Chest X-ray Segmentation System - 🔗 [GitHub](#)

Role: AI Developer | Tech: TensorFlow, U-Net, ResNet50, Computer Vision, OpenCV

- Developed an automated lung segmentation and abnormality detection model using U-Net with a ResNet50 backbone for chest X-ray analysis.
- Applied transfer learning and extensive data augmentation, achieving a Dice similarity coefficient above 0.80 across varying image conditions.
- Engineered automated highlighting features to identify potential abnormal lung regions for medical image analysis.

Shipment Price Prediction System - 🔗 [GitHub](#)

Role: ML Engineer | Tech: Python, NumPy, Pandas, Seaborn, Scikit-learn, FastAPI, Docker, AWS

- A machine learning model to predict shipment costs based on factors such as distance, weight, delivery type, shipping method, and other logistics features. Performed data preprocessing, feature engineering, model training, and hyperparameter tuning to build an accurate price prediction system.
- Developed an end-to-end machine learning system for shipment price prediction with real-time inference capabilities.
- Containerized the application using Docker and deployed the API on AWS infrastructure.

EDUCATION

Bachelor of Science (BSc)

Narsingdi Government College, Bangladesh